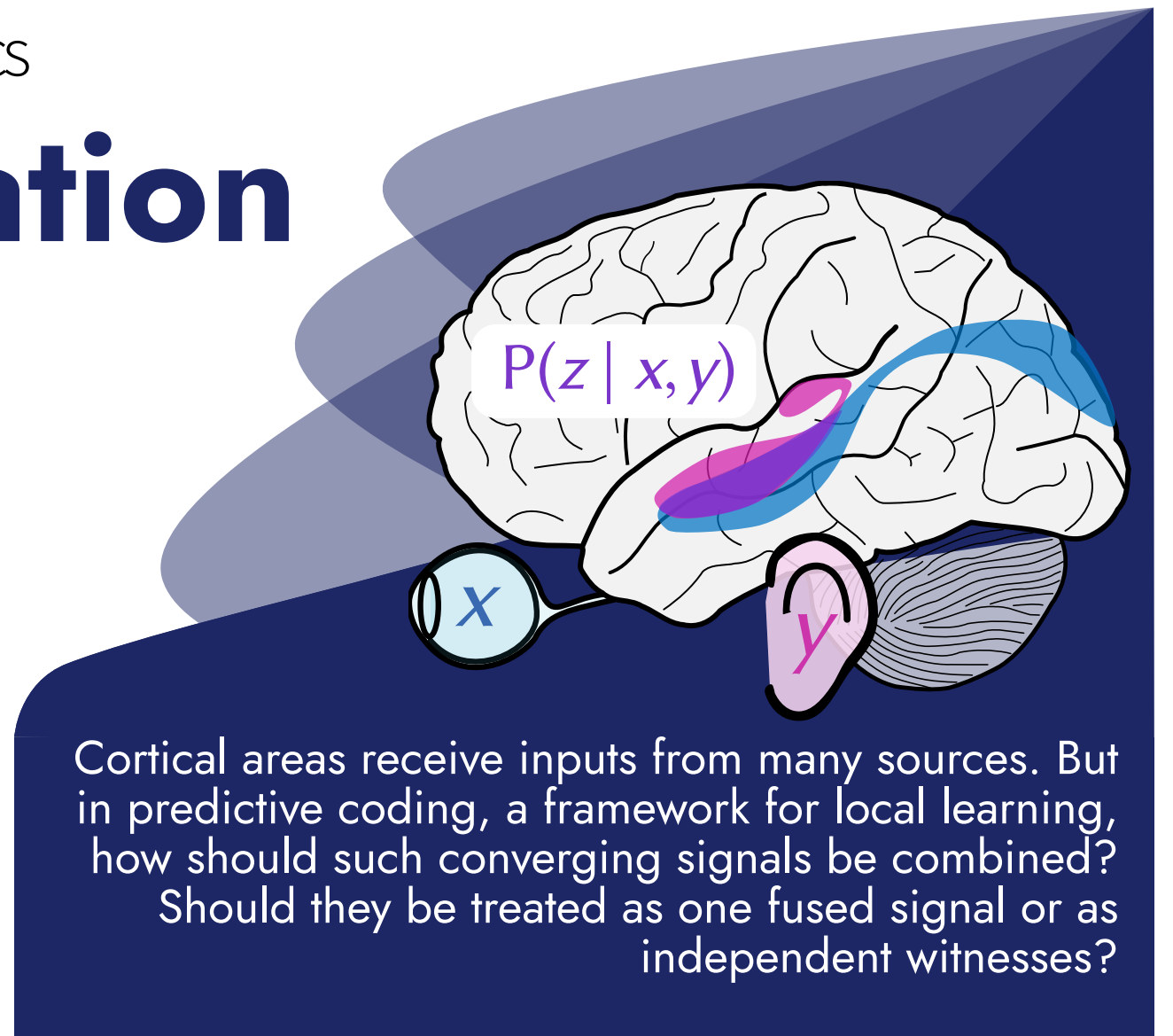
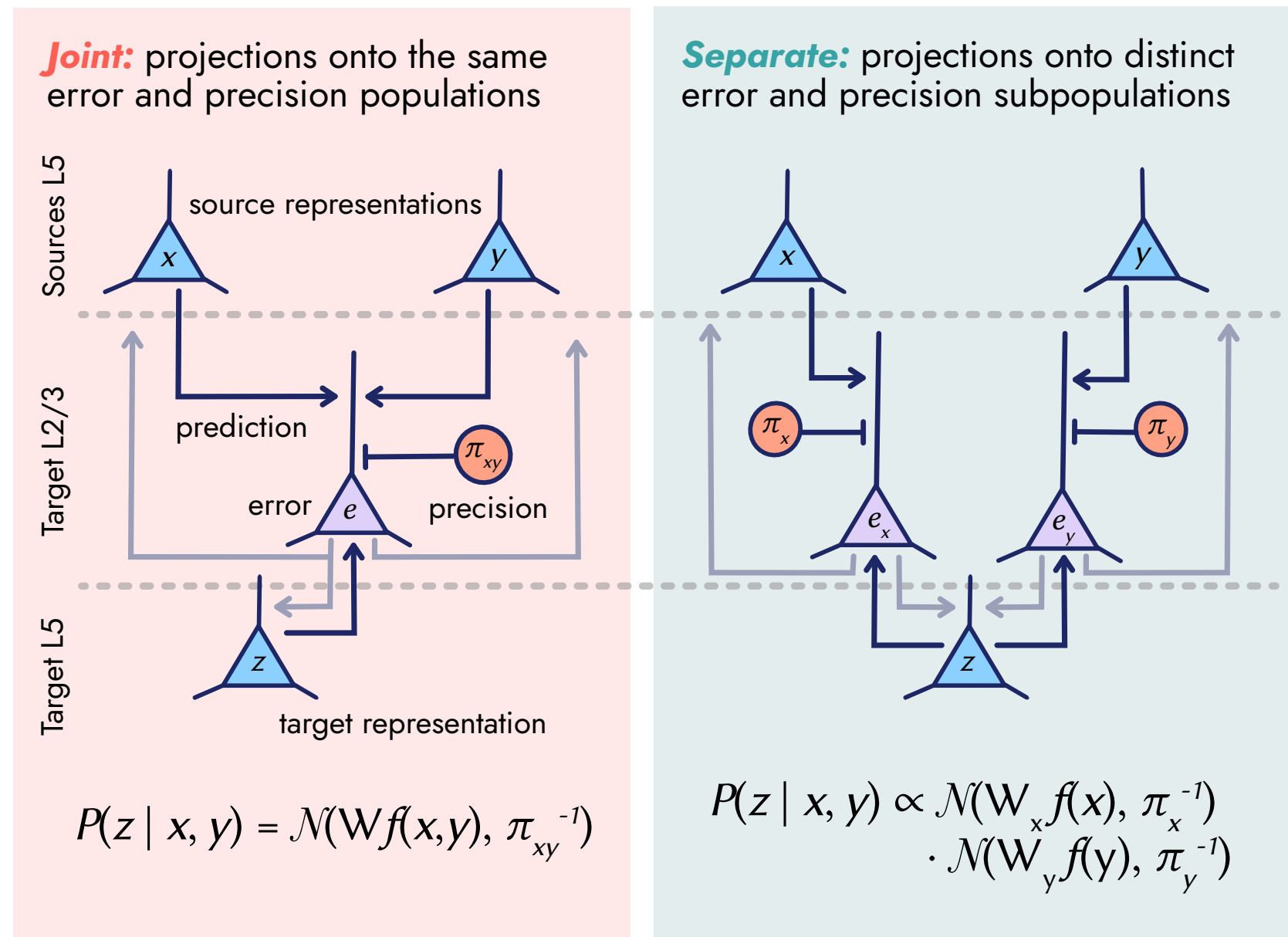


Multi-sensory Integration in Predictive Coding

Nicole Sandra-Yaffa Dumont and Katharina A. Wilmes

Predicting a latent cause: different generative models produce different circuits



When is one scheme better?

Joint for....

- Nonlinear fusion is needed
- Unsupervised learning
- 'Explaining away' effects

Separate for....

- Each stream can explain the target
- Unimodal robustness
- Per-stream precision

What do the schemes imply?

Joint means....

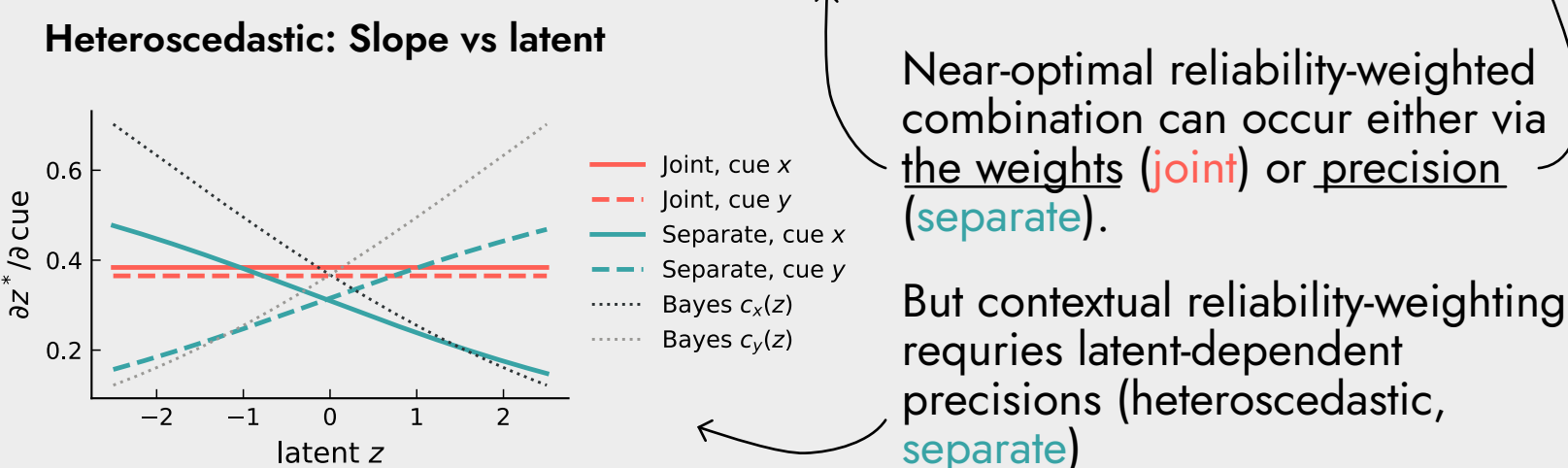
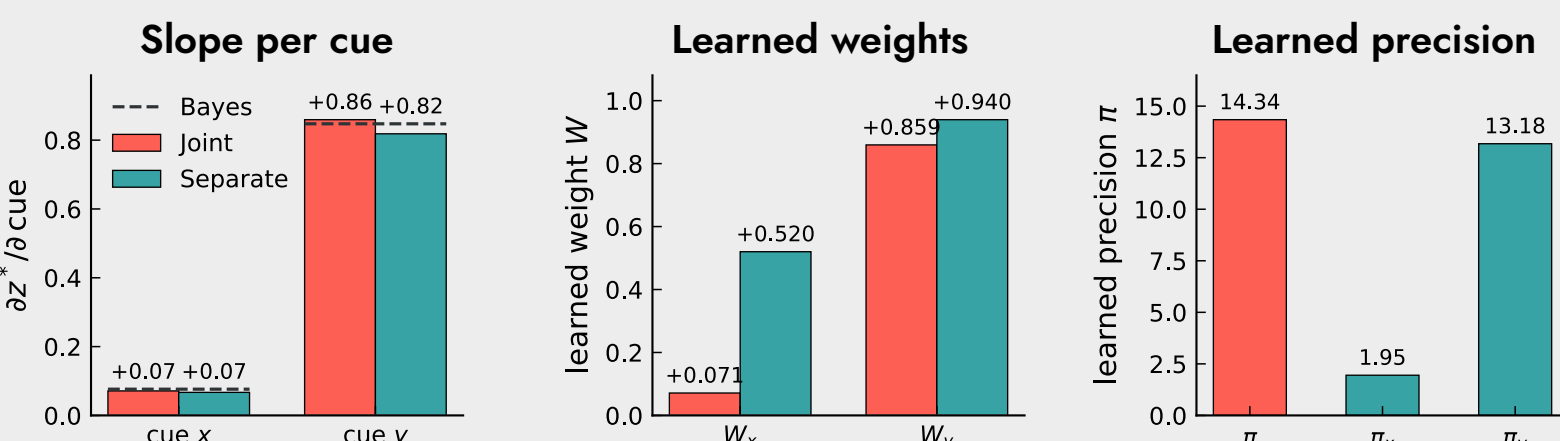
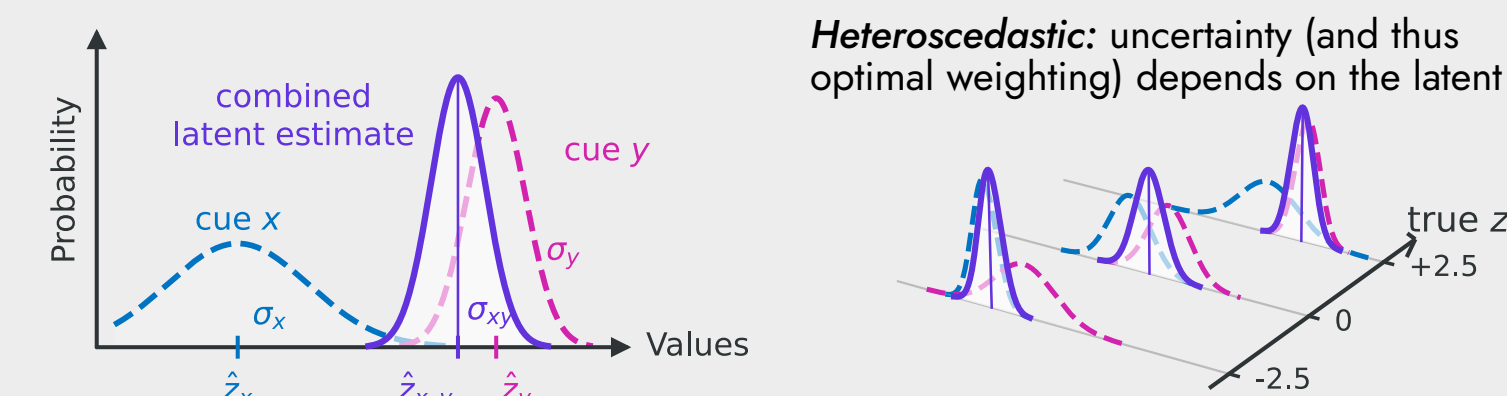
- Sensory areas project onto the same L2/3 cells in multisensory areas
- Fast cross-modal effects

Separate means....

- Projections to non-overlapping cells in multisensory areas
- Slower cross-modal effects

Optimal cue integration via two pathways

Combine two noisy cues x and y (e.g., evidence from sensory modalities) to obtain the latent z

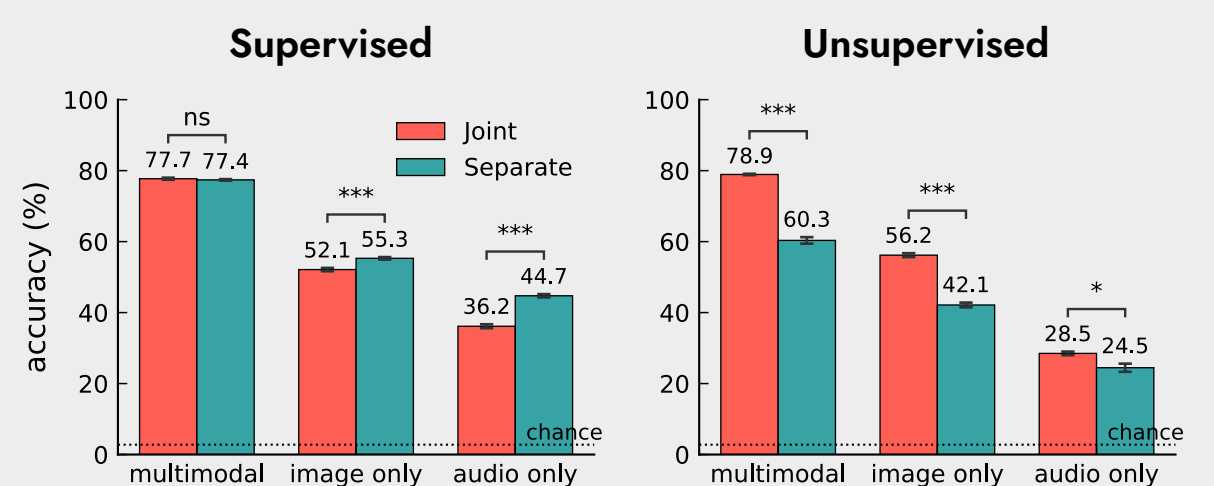
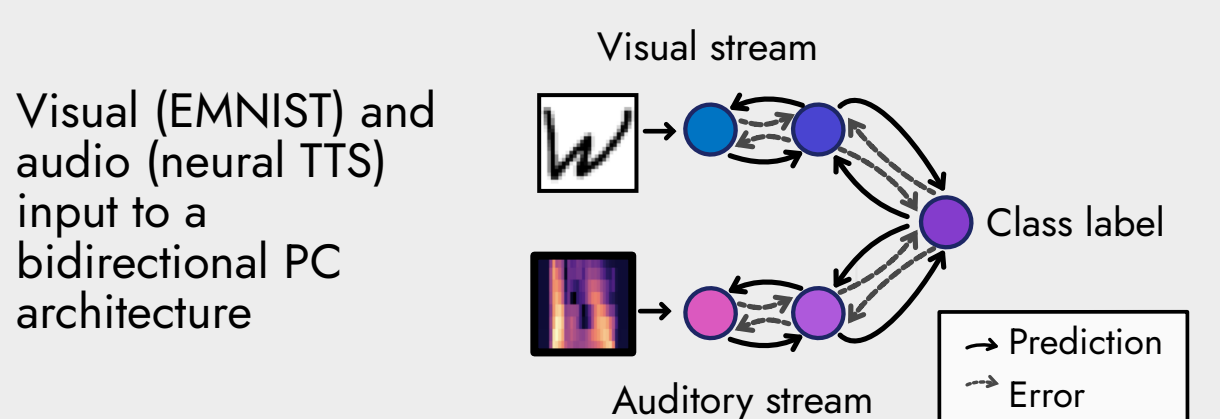


Future work

When should circuits fuse sensory sources, and when should they maintain separate predictions? Future work could explore learning rules that select or combine the two schemes, linking circuit organization to structural inference over the underlying generative model.

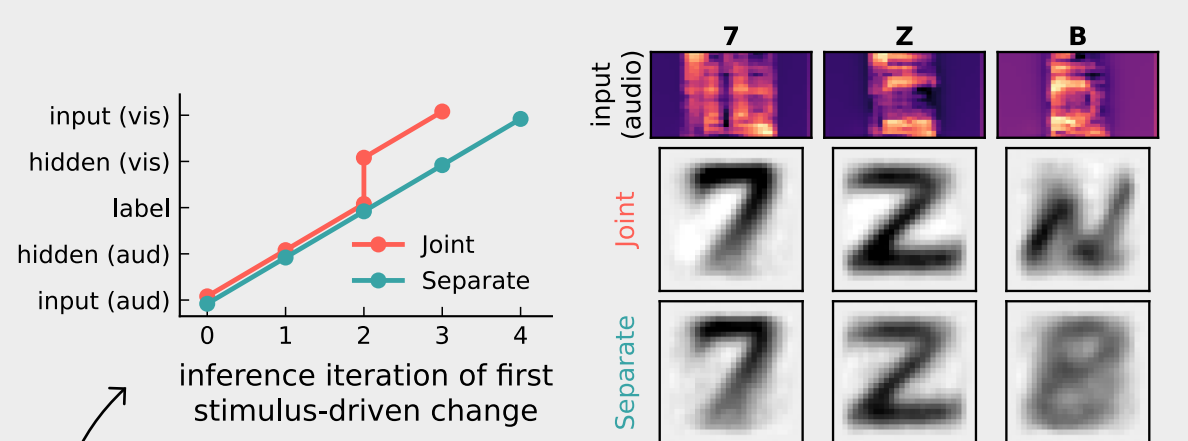
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Audiovisual tasks



With supervised training, multimodal accuracy is similar but **separate** wins on unimodal accuracy.

With purely a generative objective, **joint** results in stronger fusion and thus accuracy.



A single modality can produce predictions and errors in the other stream. Cross-modal effects have different latencies in the schemes.

Contact me at nidumon@ini.uzh.ch to learn more or collaborate!



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